Physiology: Control Mechanisms of Respiration

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Do.
Make.
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Reinvent the Future.

Objectives:

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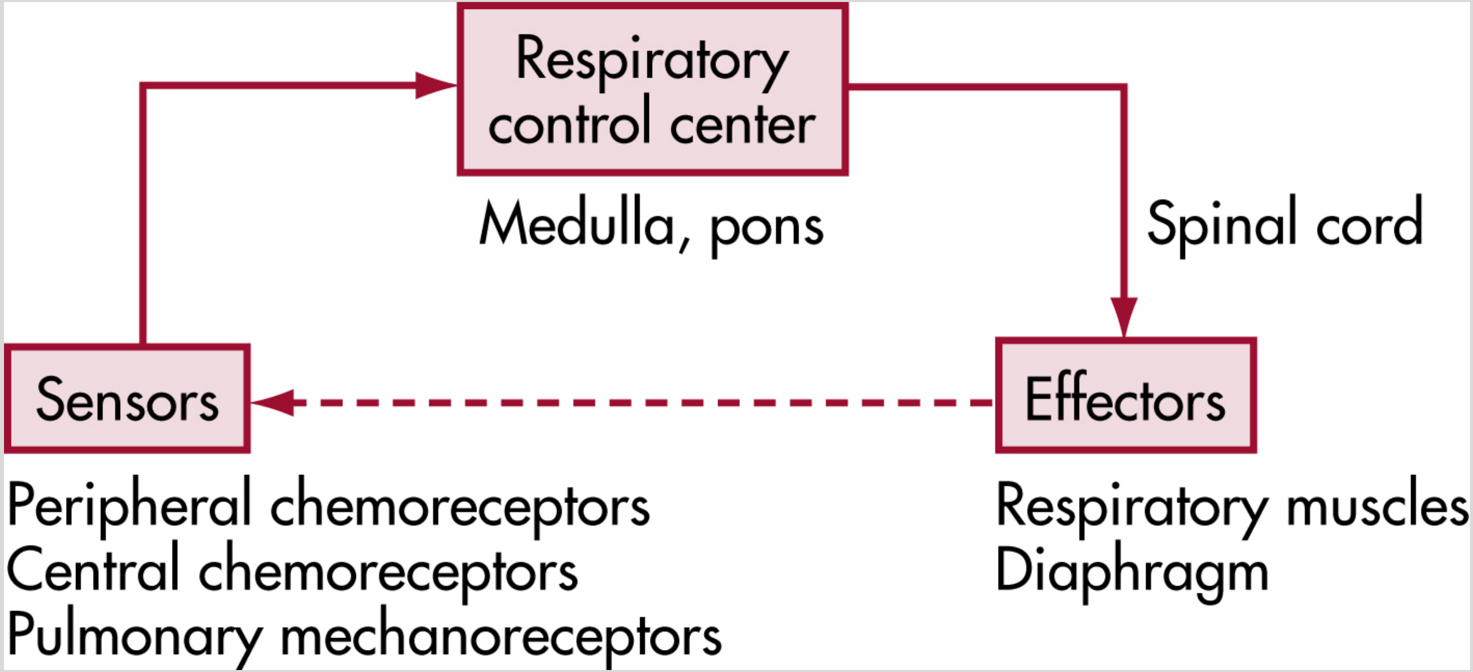
- Name the components of the feedback loop of respiratory control, including phrenic and cranial nerves, and explain how they work together as a unit.
- Name the brainstem centers in control of breathing and outline their basic functions. Compare & contrast focal vs. global hypoxia.
- Explain how the bicarbonate buffer system facilitates the work of the central chemoreceptors of breathing.
- Name the locations where peripheral chemoreceptors can be found and explain their basic characteristics.
- Name other peripheral receptors that influence breathing, what they react to, and how they affect breathing.
- Associate various breathing conditions with their breathing pattern, and potential underlying causes.
- Explain how respiratory physiology adapts to strenuous exercise.
- Explain the effects of high altitude on alveolar and arterial O_2 and how the respiratory system reacts in the short and long term.
- Compare and contrast the causes and effects of hypoxia and hypoxemia on the respiratory system.

Feedback Loop of Respiratory Control



Control of Breathing – Expanded View

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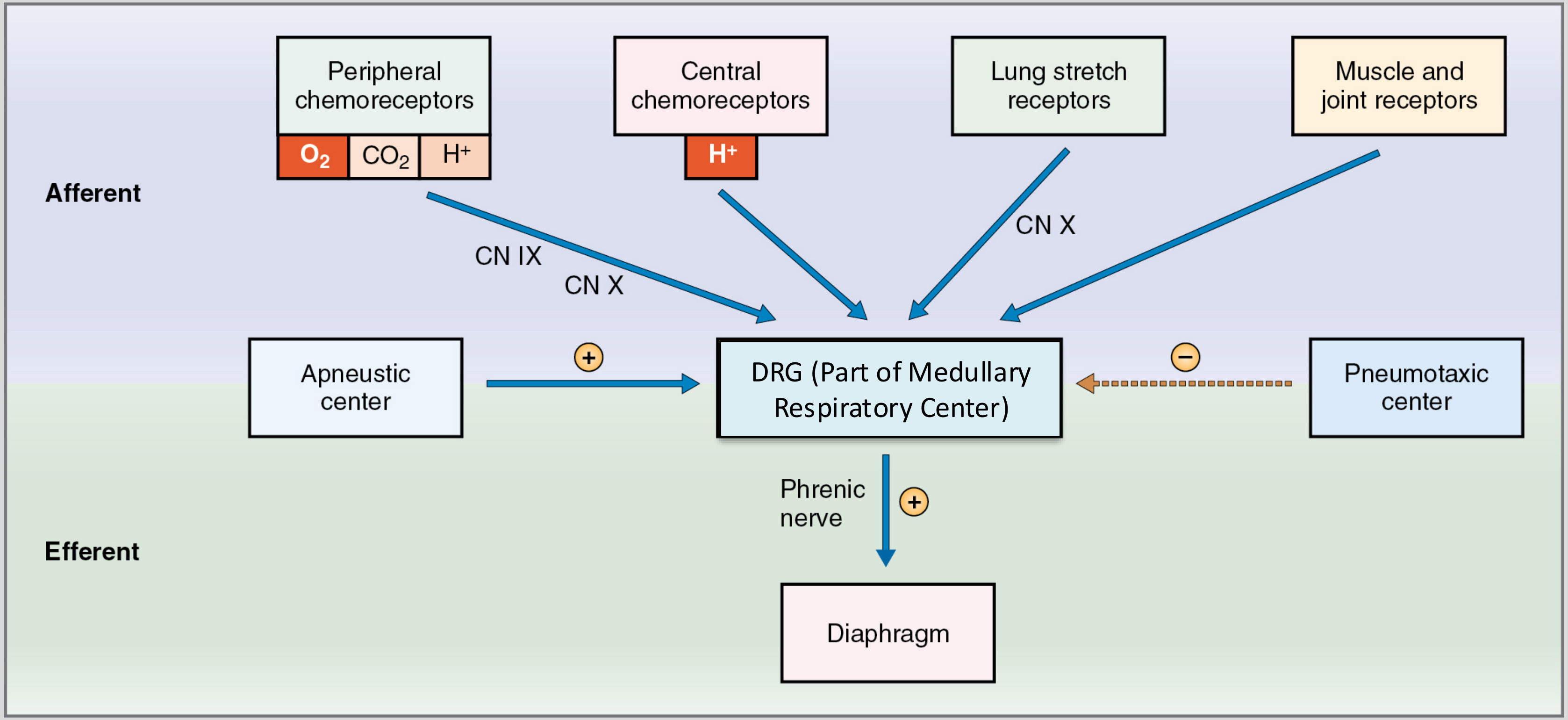

Brain Stem Centers in Control of Breathing

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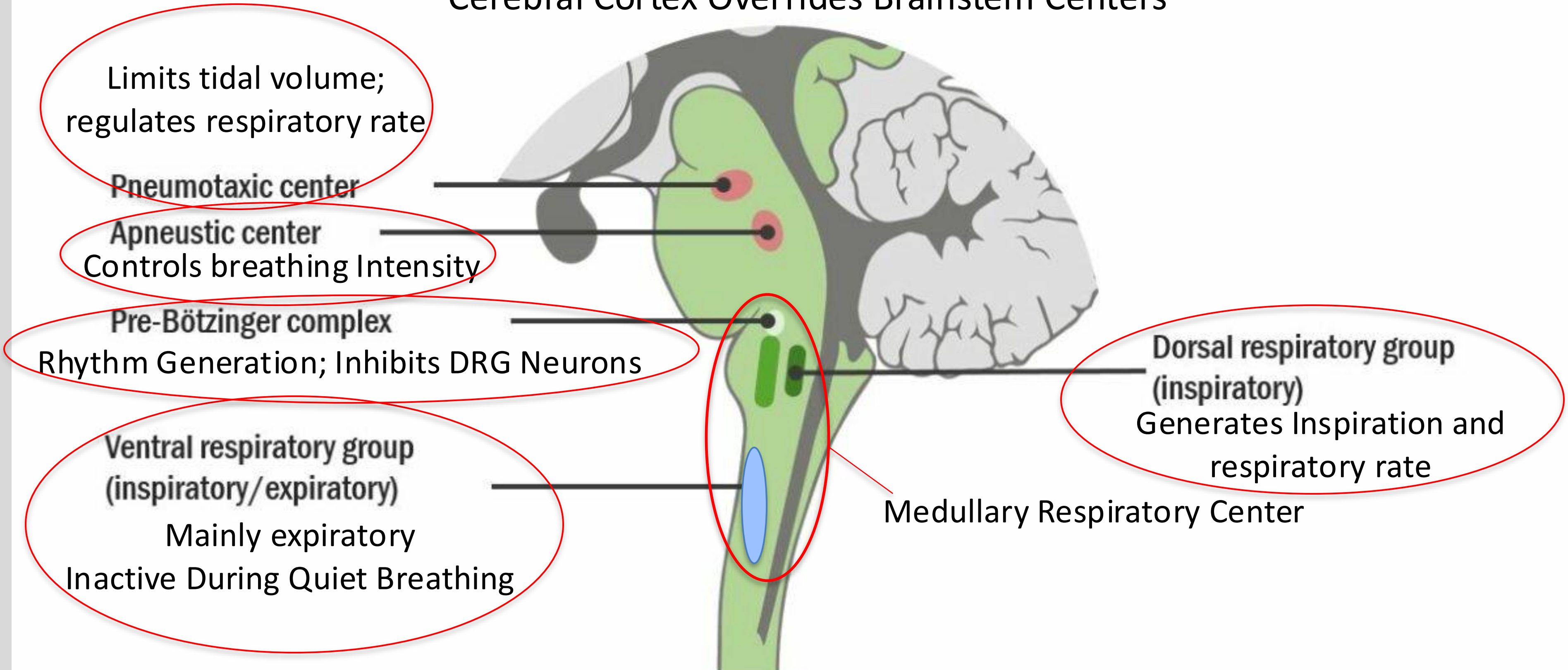
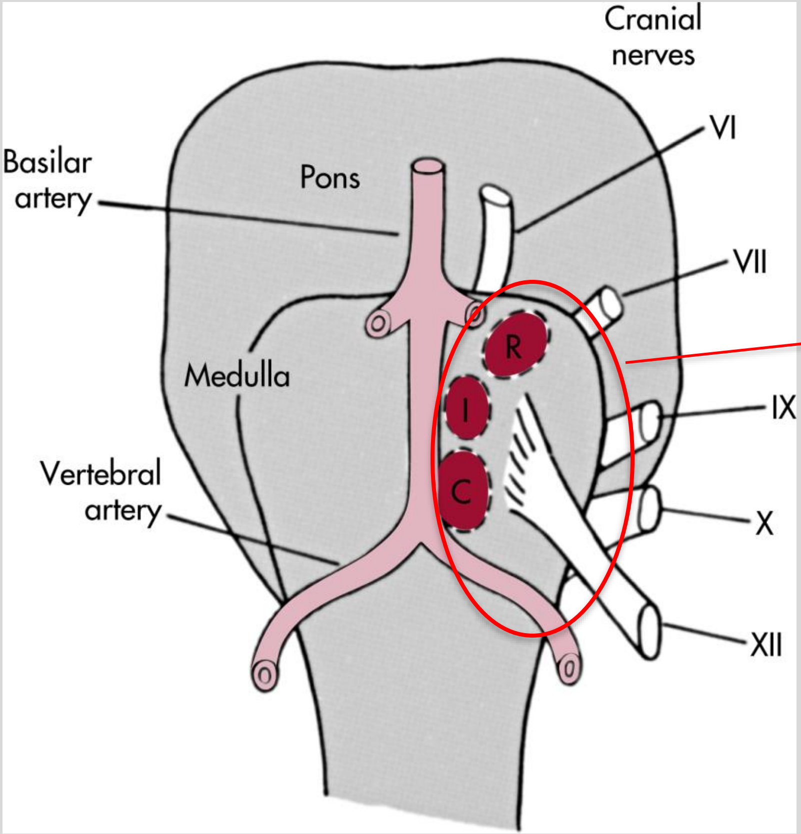
Cerebral Cortex Overrides Brainstem Centers



Central Chemoreceptors

Minute-to-Minute Breathing



Ventrolateral Medulla

R: Rostral

I: Intermediate

C: Caudate Receptor Area

DRG nearby for communication

Medullary Chemoreceptors

pH of CSF (directly); CO_2 (indirectly)

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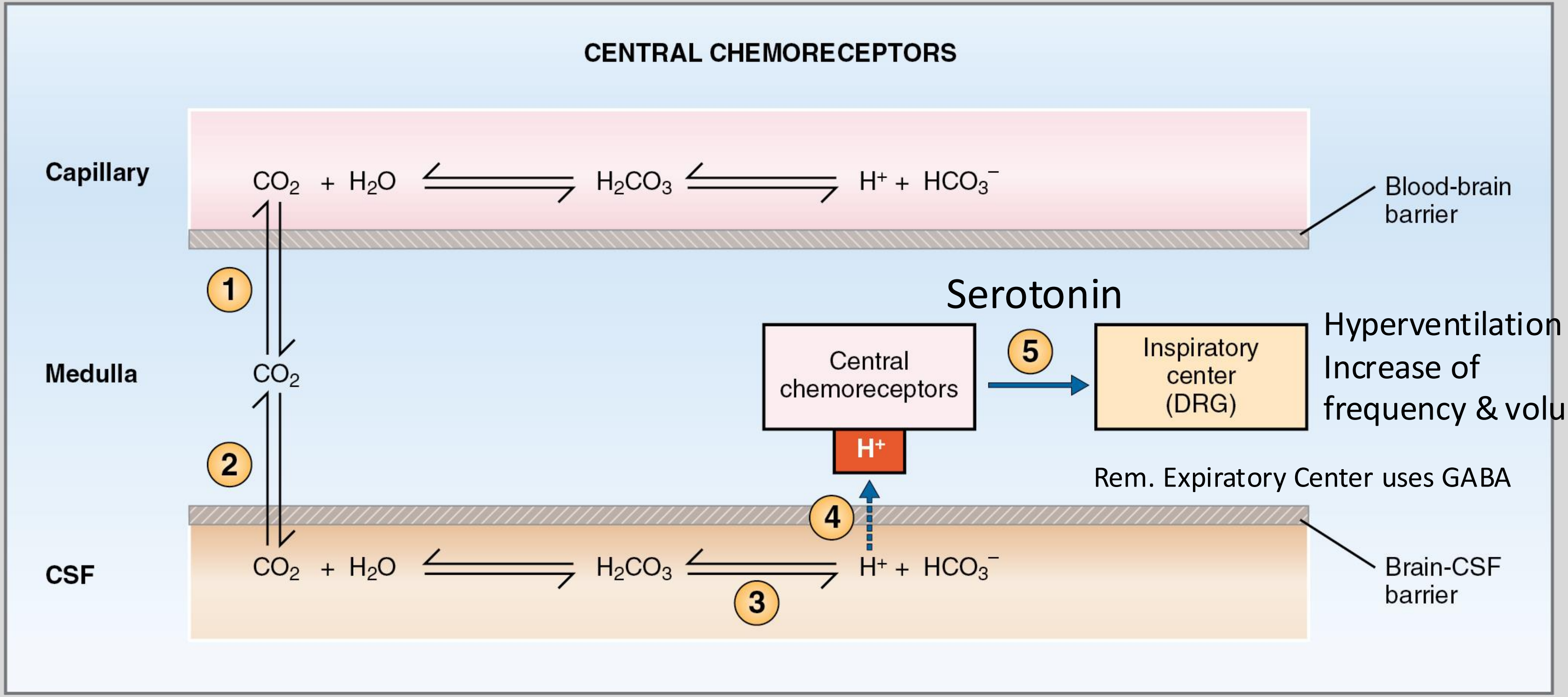

Serotonin

Hyperventilation

Increase of frequency & volume

Rem. Expiratory Center uses GABA

In sudden infant death syndrome (SIDS), there is a deficit of serotonergic neurons

Peripheral Chemoreceptors
O_2 , CO_2 , & H^+

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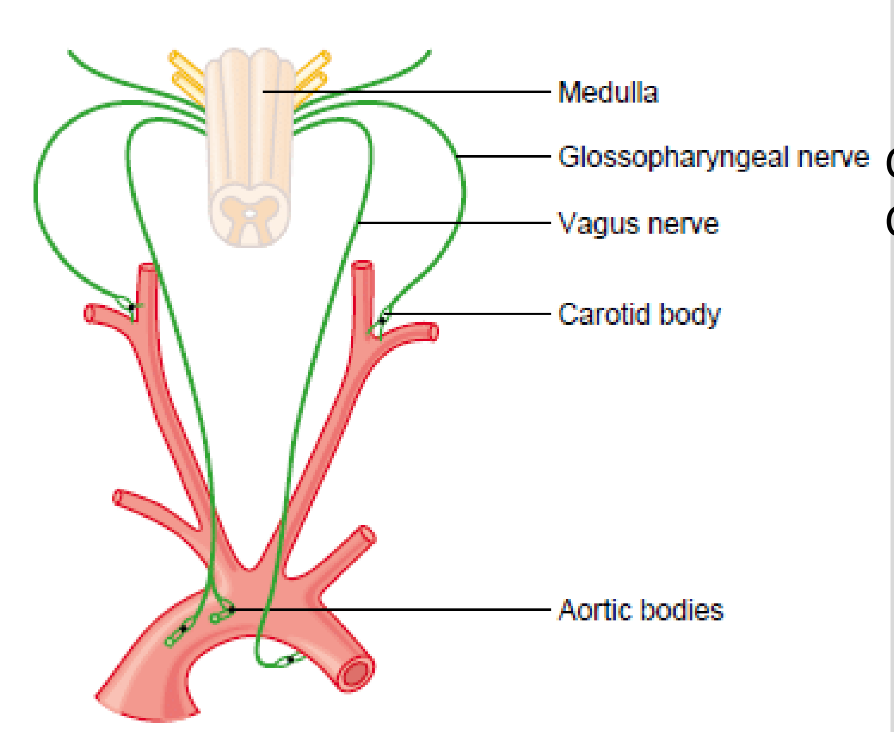

CNIX

CNX

Decreases in PaO_2 (less than 60mmHg)
Increases in $PaCO_2$ (less dominant)
Decreases in pH (carotid bodies only)

Other Peripheral Receptors

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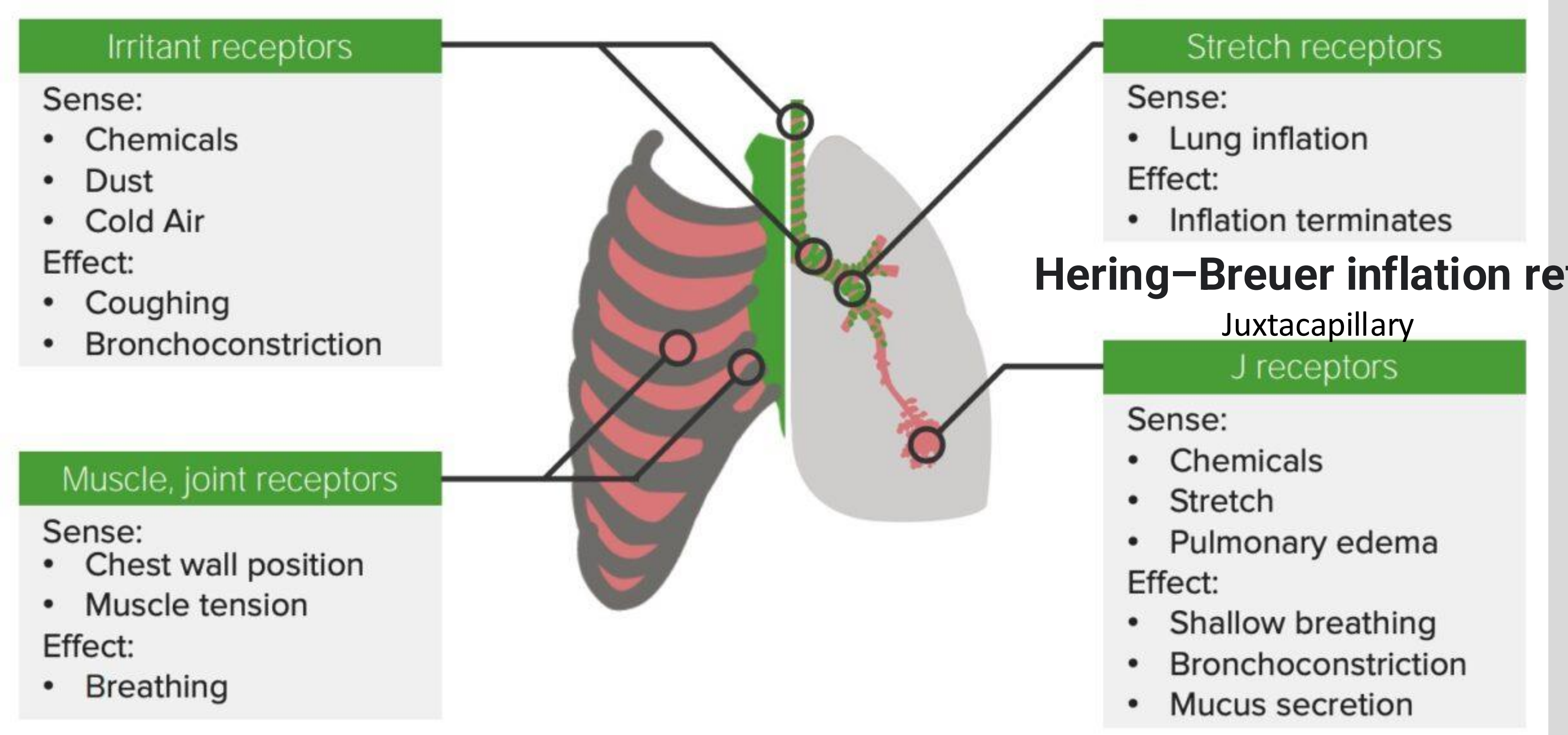

Hering–Breuer inflation reflex!

Juxtacapillary

Dyspnea in left-sided heart failure

Diagnostic Analysis of Breathing Patterns

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	Condition	Description	Causes
	Eupnea	Normal breathing rate and pattern	
	Tachypnea	Increased respiratory rate	Fever, anxiety, exercise, shock
	Bradypnea	Decreased respiratory rate	Sleep, drugs, metabolic disorder, head injury, stroke
	Apnea	Absence of breathing	Deceased patient, head injury, stroke
	Hyperpnea	Normal rate, but deep respirations	Emotional stress, diabetic ketoacidosis
	Cheyne-Stokes	Gradual increases and decreases in respirations with periods of apnea	Increasing intracranial pressure, brain stem injury
	Biot's	Rapid, deep respirations (gasps) with short pauses between sets	Spinal meningitis, many CNS causes, head injury
	Kussmaul's	Tachypnea and hyperpnea	Renal failure, metabolic acidosis, diabetic ketoacidosis
	Apneustic	Prolonged inspiratory phase with shortened expiratory phase	Lesion in brain stem

Responses to Exercise

O_2 Consumption from 250 mL/min to 4000 mL/min

Ventilation rate from 7.5 L/min to 120 L/min

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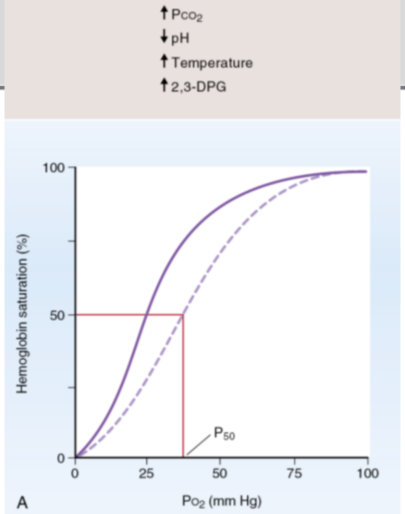

EXERCISE

↑ O_2 consumption

↑ CO_2 production

↑ Ventilation

ARTERIAL BLOOD

No change Pa_{O_2}

No change Pa_{CO_2}

No change pH (or ↓)

VENOUS BLOOD

↑ $P\bar{v}_{CO_2}$

PULMONARY BLOOD FLOW

↑ Cardiac output

↑ Pulmonary blood flow

$\dot{V}/\dot{Q}$ more *even* in lungs

O_2-HEMOGLOBIN CURVE

Shifts to right

↑ P_{50}

↓ Affinity for O_2

Adaptions to High Altitude

Mount Everest PO_2 of Inspired Air ~47 mm Hg!

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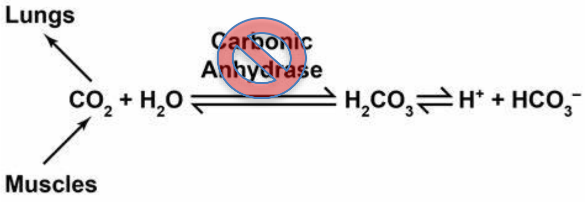
HIGH ALTITUDE

acetazolamide



↓ Patm

↓ Alveolar P_{O_2}

Causing Pulmonary Hypoxic Vasoconstriction; **Remember Why?**

Acute Altitude Sickness:
Headache, Fatigue, Dizziness, Nausea, Palpitations, & Insomnia

VENTILATION

↑ $\dot{V}_A$

Hyperventilation (initially) at PO_2 ≤ 60mmHg

ARTERIAL BLOOD

↓ Pa_{O_2}

↑ pH (respiratory alkalosis)

PULMONARY BLOOD FLOW

↑ Pulmonary resistance

↑ Pulmonary artery pressure

Hypertrophy of right ventricle

O_2-HEMOGLOBIN CURVE

↑ 2,3-DPG

Shifts to right

↑ P_{50}

↓ Affinity for O_2

ERYTHROPOIETIN

↓ Pa_{O_2}

Hypoxia

↑ Renal synthesis of EPO

Polycytemia

Hypoxemia – Decrease in PaO_2

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Cause	PaO2	A – a Gradient	Supplemental O2 Helpful?
High altitude (↓ P_b; ↓ P_{iO_2})	Decreased	Normal	Yes
Hypoventilation (↓ P_{aO_2})	Decreased	Normal	Yes
Diffusion defect (e.g., fibrosis)	Decreased	Increased	Yes
$\dot{V}/\dot{Q}$ defect	Decreased	Increased	Yes
Right-to-left shunt	Decreased	Increased	Limited
COVID-19	**Decreased**	**Increased**	**Yes**

Hypoxia – Decrease in O_2 delivery to Tissues

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Cause	Mechanism	PaO2
↓ Cardiac output	↓ Blood flow	—
Hypoxemia	↓ Pa O_2 ↓ O_2 saturation of hemoglobin ↓ O_2 content of blood	↓
Anemia	↓ Hemoglobin concentration ↓ O_2 content of blood	—
Carbon monoxide poisoning	↓ O_2 content of blood Left shift of O_2-hemoglobin curve	—
Cyanide poisoning	↓ O_2 utilization by tissues	—

Medical Board-Style Practice Question

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During an experimental trial of mechanical ventilation on an anesthetized animal, a researcher temporarily disables the ventilator's safety limits and delivers a sudden, massive volume of air that severely over-inflates the animal's lungs. The researcher observes an immediate, involuntary arrest of the animal's inspiratory effort and a prolongation of its expiratory phase. Which of the following pathways is responsible for this protective respiratory reflex?

A. J-receptors in the alveolar walls detect the stretch and signal the spinal cord to cause rapid, shallow gasping.
B. Carotid body chemoreceptors detect the sudden stretch and signal the apneustic center in the pons to promote gasping.
C. Slowly adapting pulmonary stretch receptors in the airway smooth muscle send inhibitory signals via the vagus nerve (CN X) to the medulla to terminate inspiration.
D. Irritant receptors in the trachea detect the high pressure and trigger an immediate, prolonged inspiratory gasp to protect the alveoli.
E. Central chemoreceptors in the fourth ventricle sense the sudden lung inflation and directly block phrenic nerve firing at the spinal level.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

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